Chapter 6

The Laplace transform

6.1 The Laplace transform

Note: 1.5–2 lectures, §7.1 in [EP], §6.1 and parts of §6.2 in [BD]

6.1.1 The transform

In this chapter, we will discuss the Laplace transform[†]. The Laplace transform is a very efficient method to solve certain ODE or PDE problems. The transform takes a differential equation and turns it into an algebraic equation. If the algebraic equation can be solved, applying the inverse transform gives us our desired solution. The Laplace transform also has applications in the analysis of electrical circuits, NMR spectroscopy, signal processing, and elsewhere. Finally, understanding the Laplace transform will also help with understanding the related Fourier transform, which, however, requires more understanding of complex numbers. We will not cover the Fourier transform.

The Laplace transform also gives a lot of insight into the nature of the equations we are dealing with. It can be seen as converting between the time and the frequency domain. For example, take the standard equation

$$mx''(t) + cx'(t) + kx(t) = f(t).$$

We can think of t as time and $f(t)$ as incoming signal. The Laplace transform will convert the equation from a differential equation in time to an algebraic (no derivatives) equation, where the new independent variable s is thought of as the frequency.[‡]

We can think of the *Laplace transform* as a black box. It eats functions and spits out functions in a new variable. We write $\mathcal{L}\{f(t)\} = F(s)$ for the Laplace transform of $f(t)$. It is common to write lower case letters for functions in the time domain and upper case letters for functions in the frequency domain. We use the same letter to denote that one

[†]Just like the Laplace equation and the Laplacian, the Laplace transform is also named after Pierre-Simon, marquis de Laplace (1749–1827).

[‡]Really, it is the "frequency" in terms of complex numbers, but we digress.

function is the Laplace transform of the other. For example, $F(s)$ is the Laplace transform of $f(t)$. Let us define the transform.

$$\boxed{\mathcal{L}\{f(t)\} = F(s) \stackrel{\text{def}}{=} \int_0^\infty e^{-st} f(t)\, dt.}$$

We note that we are only considering $t \geq 0$ in the transform. Of course, if we think of t as time, there is no problem; we are generally interested in finding out what will happen in the future (the Laplace transform is one place where it is safe to ignore the past). Let us compute some simple transforms.

Example 6.1.1: Suppose $f(t) = 1$, then

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\, dt = \left[\frac{e^{-st}}{-s}\right]_{t=0}^\infty = \lim_{h\to\infty} \left[\frac{e^{-st}}{-s}\right]_{t=0}^h = \lim_{h\to\infty} \left(\frac{e^{-sh}}{-s} - \frac{1}{-s}\right) = \frac{1}{s}.$$

The limit (the improper integral) only exists if $s > 0$. So $\mathcal{L}\{1\}$ is only defined for $s > 0$.

Example 6.1.2: Suppose $f(t) = e^{-at}$, then

$$\mathcal{L}\{e^{-at}\} = \int_0^\infty e^{-st} e^{-at}\, dt = \int_0^\infty e^{-(s+a)t}\, dt = \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_{t=0}^\infty = \frac{1}{s+a}.$$

The limit only exists if $s + a > 0$. So $\mathcal{L}\{e^{-at}\}$ is only defined for $s + a > 0$.

Example 6.1.3: Suppose $f(t) = t$, then using integration by parts

$$\begin{aligned} \mathcal{L}\{t\} &= \int_0^\infty e^{-st} t\, dt \\ &= \left[\frac{-te^{-st}}{s}\right]_{t=0}^\infty + \frac{1}{s}\int_0^\infty e^{-st}\, dt \\ &= 0 + \frac{1}{s}\left[\frac{e^{-st}}{-s}\right]_{t=0}^\infty \\ &= \frac{1}{s^2}. \end{aligned}$$

Again, the limit only exists if $s > 0$.

Example 6.1.4: A common function is the *unit step function*, which is sometimes called the *Heaviside function**. This function is generally given as

$$u(t) = \begin{cases} 0 & \text{if } \ t < 0, \\ 1 & \text{if } \ t \geq 0. \end{cases}$$

*The function is named after the English mathematician, engineer, and physicist Oliver Heaviside (1850–1925). Only by coincidence is the function "heavy" on "one side."

Let us find the Laplace transform of $u(t-a)$, where $a \geq 0$ is some constant. That is, it is the function that is 0 for $t < a$ and 1 for $t \geq a$.

$$\mathcal{L}\{u(t-a)\} = \int_0^\infty e^{-st} u(t-a)\,dt = \int_a^\infty e^{-st}\,dt = \left[\frac{e^{-st}}{-s}\right]_{t=a}^\infty = \frac{e^{-as}}{s},$$

where of course $s > 0$ (and $a \geq 0$ as we said before).

By applying similar procedures, we can compute the transforms of many elementary functions. Many basic transforms are listed in Table 6.1 (see also appendix B).

$f(t)$	$\mathcal{L}\{f(t)\}$	$f(t)$	$\mathcal{L}\{f(t)\}$
C	$\frac{C}{s}$	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
t	$\frac{1}{s^2}$	$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
t^2	$\frac{2}{s^3}$	$\sinh(\omega t)$	$\frac{\omega}{s^2-\omega^2}$
t^3	$\frac{6}{s^4}$	$\cosh(\omega t)$	$\frac{s}{s^2-\omega^2}$
t^n	$\frac{n!}{s^{n+1}}$	$u(t-a) \quad (a \geq 0)$	$\frac{e^{-as}}{s}$
e^{-at}	$\frac{1}{s+a}$		

Table 6.1: Some Laplace transforms (C, ω, and a are constants).

***Exercise* 6.1.1:** *Verify Table 6.1.*

Since the transform is defined by an integral, we can use the linearity properties of the integral. For example, if C is a constant, then

$$\mathcal{L}\{Cf(t)\} = \int_0^\infty e^{-st} Cf(t)\,dt = C\int_0^\infty e^{-st} f(t)\,dt = C\mathcal{L}\{f(t)\}.$$

So we can "pull out" a constant from the transform. Similarly, with addition. As linearity is important, we state it as a theorem.

Theorem 6.1.1 (Linearity of the Laplace transform)**.** *If A, B, and C are constants, then*

$$\boxed{\mathcal{L}\{Af(t) + Bg(t)\} = A\mathcal{L}\{f(t)\} + B\mathcal{L}\{g(t)\},}$$

and in particular,

$$\mathcal{L}\{Cf(t)\} = C\mathcal{L}\{f(t)\}.$$

***Exercise* 6.1.2:** *Verify the theorem. That is, show that $\mathcal{L}\{Af(t) + Bg(t)\} = A\mathcal{L}\{f(t)\} + B\mathcal{L}\{g(t)\}$.*

These rules together with Table 6.1 on the previous page make it easy to find the Laplace transform of a whole lot of functions already. For example:

$$\mathcal{L}\left\{2 + 5t + 9e^{-2t}\right\} = \mathcal{L}\left\{2\right\} + 5\mathcal{L}\left\{t\right\} + 9\mathcal{L}\left\{e^{-2t}\right\} = \frac{2}{s} + \frac{5}{s^2} + \frac{9}{s+2}.$$

Be careful! The Laplace transform of a product is *not* the product of the transforms. In general

$$\mathcal{L}\left\{f(t)g(t)\right\} \neq \mathcal{L}\left\{f(t)\right\}\mathcal{L}\left\{g(t)\right\}.$$

Moreover, not all functions have a Laplace transform. For example, the function $\frac{1}{t}$ does not have a Laplace transform as the integral diverges for all s. Similarly, $\tan t$ or e^{t^2} do not have Laplace transforms.

6.1.2 Existence and uniqueness

When does the Laplace transform exist? A function $f(t)$ is of *exponential order* as t goes to infinity if

$$|f(t)| \leq Me^{ct},$$

for some constants M and c, for sufficiently large t (say for all $t > t_0$ for some t_0). The simplest way to check this condition is to try and compute

$$\lim_{t\to\infty} \frac{f(t)}{e^{ct}}.$$

If the limit exists and is finite (usually zero), then $f(t)$ is of exponential order.

***Exercise* 6.1.3:** *Use L'Hopital's rule from calculus to show that a polynomial is of exponential order. Hint: Note that a sum of two exponential-order functions is also of exponential order. Then show that t^n is of exponential order for any n.*

For an exponential-order function, we have existence and uniqueness of the Laplace transform.

Theorem 6.1.2 (Existence)**.** *Let $f(t)$ be continuous on the interval $[0,\infty)$ and of exponential order for a certain constant c. Then $F(s) = \mathcal{L}\left\{f(t)\right\}$ is defined for all $s > c$.*

The existence is not difficult to see. Let $f(t)$ be of exponential order, that is, $|f(t)| \leq Me^{ct}$ for all $t > 0$ (for simplicity $t_0 = 0$). Let $s > c$, or in other words $(s - c) > 0$. By the comparison theorem from calculus, the improper integral defining $\mathcal{L}\left\{f(t)\right\}$ exists because the following integral exists

$$\int_0^\infty e^{-st}(Me^{ct})\,dt = M\int_0^\infty e^{-(s-c)t}\,dt = M\left[\frac{e^{-(s-c)t}}{-(s-c)}\right]_{t=0}^\infty = \frac{M}{s-c}.$$

The transform also exists for some other functions that are not of exponential order, but that will not be relevant to us. Before dealing with uniqueness, we note that for functions of exponential order, their Laplace transform decays at infinity:

$$\lim_{s\to\infty} F(s) = 0.$$

Theorem 6.1.3 (Uniqueness)**.** *Let $f(t)$ and $g(t)$ be continuous and of exponential order. Suppose that there exists a constant C, such that $F(s) = G(s)$ for all $s > C$. Then $f(t) = g(t)$ for all $t \geq 0$.*

Both theorems hold for piecewise continuous functions as well. Recall that piecewise continuous means that the function is continuous except perhaps at a discrete set of points, where it has jump discontinuities like the Heaviside function. The Laplace transform, however, does not "see" values at the discontinuities. So we can only conclude that $f(t) = g(t)$ outside of discontinuities. For example, the unit step function is sometimes defined using $u(0) = 1/2$. This new step function, however, has the exact same Laplace transform as the one we defined earlier, where $u(0) = 1$.

6.1.3 The inverse transform

As we said, the Laplace transform will allow us to convert a differential equation into an algebraic equation. Once we solve the algebraic equation in the frequency domain, we will want to get back to the time domain, as that is what we are interested in. Given a function $F(s)$, we wish to find a function $f(t)$ such that $\mathcal{L}\{f(t)\} = F(s)$. Theorem 6.1.3 says that the solution $f(t)$ is unique. So we can without fear make the following definition.

Suppose $F(s) = \mathcal{L}\{f(t)\}$ for some function $f(t)$. Define the *inverse Laplace transform* as

$$\mathcal{L}^{-1}\{F(s)\} \overset{\text{def}}{=} f(t).$$

There is an integral formula for the inverse, but it is not as simple as the transform itself—it requires complex numbers and path integrals. For us it will suffice to compute the inverse using Table 6.1 on page 295.

Example 6.1.5: Take $F(s) = \frac{1}{s+1}$. Find the inverse Laplace transform.

We look at the table to find

$$\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}.$$

As the Laplace transform is linear, the inverse Laplace transform is also linear. That is,

$$\mathcal{L}^{-1}\{AF(s) + BG(s)\} = A\mathcal{L}^{-1}\{F(s)\} + B\mathcal{L}^{-1}\{G(s)\}.$$

Of course, we also have $\mathcal{L}^{-1}\{AF(s)\} = A\mathcal{L}^{-1}\{F(s)\}$.

Example 6.1.6: Take $F(s) = \frac{s^2+s+1}{s^3+s}$. Find the inverse Laplace transform.

First we use the *method of partial fractions* to write F in a form where we can use Table 6.1 on page 295. We factor the denominator as $s(s^2 + 1)$ and write

$$\frac{s^2 + s + 1}{s^3 + s} = \frac{A}{s} + \frac{Bs + C}{s^2 + 1}.$$

Putting the right-hand side over a common denominator and equating the numerators, we get $A(s^2 + 1) + s(Bs + C) = s^2 + s + 1$. Expanding and equating coefficients, we obtain

$A + B = 1$, $C = 1$, $A = 1$, and thus $B = 0$. In other words,

$$F(s) = \frac{s^2 + s + 1}{s^3 + s} = \frac{1}{s} + \frac{1}{s^2 + 1}.$$

By linearity of the inverse Laplace transform, we get

$$\mathcal{L}^{-1}\left\{\frac{s^2 + s + 1}{s^3 + s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\} = 1 + \sin t.$$

Another useful property is the so-called *shifting property* or the *first shifting property*

$$\boxed{\mathcal{L}\left\{e^{-at} f(t)\right\} = F(s + a),}$$

where $F(s)$ is the Laplace transform of $f(t)$.

***Exercise* 6.1.4:** *Derive the first shifting property from the definition of the Laplace transform.*

The shifting property can be used, for example, when the denominator is a more complicated quadratic that may come up in the method of partial fractions. We complete the square and write such quadratics as $(s + a)^2 + b$ and then use the shifting property.

Example 6.1.7: Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2+4s+8}\right\}$.

First, we complete the square to make the denominator $(s + 2)^2 + 4$. We find

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2 + 2^2}\right\} = \frac{1}{2}\sin(2t).$$

Finally, we put it all together with the shifting property to find

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4s + 8}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s + 2)^2 + 4}\right\} = \frac{1}{2}e^{-2t}\sin(2t).$$

Often, we want to be able to apply the inverse Laplace transform to rational functions, that is, functions of the form

$$\frac{F(s)}{G(s)}$$

where $F(s)$ and $G(s)$ are polynomials. If $\frac{F(s)}{G(s)}$ is the Laplace transform of an exponential-order function, it goes to zero as $s \to \infty$, and so the degree of $F(s)$ must be smaller than that of $G(s)$. Such rational functions are called *proper rational functions* and we can always apply the method of partial fractions without polynomial division. Of course, we still need to be able to factor the denominator into linear and quadratic terms, which involves finding the roots of the denominator.

6.1.4 Exercises

Exercise* 6.1.5*: *Find the Laplace transform of* $3 + t^5 + \sin(\pi t)$.

Exercise* 6.1.6*: *Find the Laplace transform of* $a + bt + ct^2$ *for some constants* a, b, *and* c.

Exercise* 6.1.7*: *Find the Laplace transform of* $A\cos(\omega t) + B\sin(\omega t)$.

Exercise* 6.1.8*: *Find the Laplace transform of* $\cos^2(\omega t)$.

Exercise* 6.1.9*: *Find the inverse Laplace transform of* $\frac{4}{s^2-9}$.

Exercise* 6.1.10*: *Find the inverse Laplace transform of* $\frac{2s}{s^2-1}$.

Exercise* 6.1.11*: *Find the inverse Laplace transform of* $\frac{1}{(s-1)^2(s+1)}$.

Exercise* 6.1.12*: *Find the Laplace transform of* $f(t) = \begin{cases} t & \text{if } t \geq 1, \\ 0 & \text{if } t < 1. \end{cases}$

Exercise* 6.1.13*: *Find the inverse Laplace transform of* $\frac{s}{(s^2+s+2)(s+4)}$.

Exercise* 6.1.14*: *Find the Laplace transform of* $\sin\big(\omega(t-a)\big)$.

Exercise* 6.1.15*: *Find the Laplace transform of* $t\sin(\omega t)$. *Hint: Several integrations by parts.*

Exercise* 6.1.101*: *Find the Laplace transform of* $4(t+1)^2$.

Exercise* 6.1.102*: *Find the inverse Laplace transform of* $\frac{8}{s^3(s+2)}$.

Exercise* 6.1.103*: *Find the Laplace transform of* te^{-t}. *Hint: Shifting property.*

Exercise* 6.1.104*: *Find the Laplace transform of* $\sin(t)e^{-t}$. *Hint: Integrate by parts.*

6.2 Transforms of derivatives and ODEs

Note: 2 lectures, §7.2–7.3 in [EP], §6.2 and §6.3 in [BD]

6.2.1 Transforms of derivatives

Let us see how the Laplace transform is used for differential equations. First we find the Laplace transform of the derivative of a function. Suppose $g(t)$ is a differentiable function of exponential order, that is, $|g(t)| \leq Me^{ct}$ for some M and c. So $\mathcal{L}\big\{g(t)\big\}$ exists, and what is more, $\lim_{t\to\infty} e^{-st} g(t) = 0$ when $s > c$. Then

$$\mathcal{L}\big\{g'(t)\big\} = \int_0^\infty e^{-st} g'(t)\, dt = \Big[e^{-st} g(t)\Big]_{t=0}^{\infty} - \int_0^\infty (-s)\, e^{-st} g(t)\, dt = -g(0) + s\mathcal{L}\big\{g(t)\big\}.$$

We repeat this procedure for higher derivatives. The results are listed in Table 6.2. The procedure also works for continuous piecewise smooth functions, that is, functions that are continuous with a piecewise continuous derivative.

$f(t)$	$\mathcal{L}\big\{f(t)\big\} = F(s)$
$g'(t)$	$sG(s) - g(0)$
$g''(t)$	$s^2G(s) - sg(0) - g'(0)$
$g'''(t)$	$s^3G(s) - s^2g(0) - sg'(0) - g''(0)$

Table 6.2: Laplace transforms of derivatives ($G(s) = \mathcal{L}\big\{g(t)\big\}$ as usual).

***Exercise* 6.2.1:** *Verify Table 6.2.*

6.2.2 Solving ODEs with the Laplace transform

Notice that the Laplace transform turns differentiation into multiplication by s. It is this property that makes it useful to apply the transform to differential equations.

Example 6.2.1: Consider the problem

$$x''(t) + x(t) = \cos(2t), \qquad x(0) = 0, \qquad x'(0) = 1.$$

We will take the Laplace transform of both sides of the equation. By $X(s)$, we will, as usual, denote the Laplace transform of $x(t)$.

$$\begin{aligned} \mathcal{L}\big\{x''(t) + x(t)\big\} &= \mathcal{L}\big\{\cos(2t)\big\}, \\ s^2X(s) - sx(0) - x'(0) + X(s) &= \frac{s}{s^2+4}. \end{aligned}$$

We plug in the initial conditions now—making computations more streamlined—to find

$$s^2X(s) - 1 + X(s) = \frac{s}{s^2+4}.$$

We solve for $X(s)$,

$$X(s) = \frac{s}{(s^2+1)(s^2+4)} + \frac{1}{s^2+1}.$$

We use partial fractions (exercise) to write

$$X(s) = \frac{1}{3}\frac{s}{s^2+1} - \frac{1}{3}\frac{s}{s^2+4} + \frac{1}{s^2+1}.$$

Now take the inverse Laplace transform to obtain

$$x(t) = \frac{1}{3}\cos(t) - \frac{1}{3}\cos(2t) + \sin(t).$$

The procedure for linear constant-coefficient equations is as follows: Take an ordinary differential equation in the time variable t. Apply the Laplace transform to transform the equation into an algebraic (non differential) equation in the frequency domain. All the $x(t)$, $x'(t)$, $x''(t)$, and so on, will be converted to $X(s)$, $sX(s) - x(0)$, $s^2X(s) - sx(0) - x'(0)$, and so on. Solve the equation for $X(s)$. Then taking the inverse transform, if possible, find $x(t)$.

It should be noted that since not every function has a Laplace transform, not every equation can be solved in this manner. Also if the equation is not a linear constant-coefficient ODE, then applying the Laplace transform may not give us an algebraic equation.

6.2.3 Using the Heaviside function

Before we move on to more general equations than those we could solve before, we want to consider the Heaviside function. See Figure 6.1 on the following page for the graph.

$$u(t) = \begin{cases} 0 & \text{if } t < 0, \\ 1 & \text{if } t \geq 0. \end{cases}$$

The Heaviside function is useful for putting other functions together or cutting functions off. Usually we use $u(t-a)$ for some constant a; we just shift the graph to the right by a. That is, $u(t-a) = 0$ when $t < a$ and $u(t-a) = 1$ when $t \geq a$. For example, suppose $f(t)$ is a "signal", and we started receiving $\sin t$ at time $t = \pi$. The function $f(t)$ is then defined as

$$f(t) = \begin{cases} 0 & \text{if } t < \pi, \\ \sin t & \text{if } t \geq \pi. \end{cases}$$

Using the Heaviside function, $f(t)$ can be written as

$$f(t) = u(t-\pi)\sin t.$$

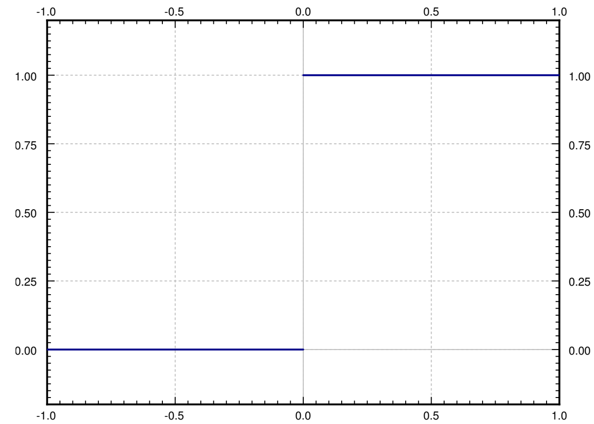

Figure 6.1: Plot of the Heaviside (unit step) function $u(t)$.

Similarly, the step function that is 1 on the interval $[1, 2)$ and 0 elsewhere is written as

$$u(t-1) - u(t-2).$$

With the Heaviside function we can express functions defined piecewise. If $f(t) = t$ when t is in $[0, 1]$, $f(t) = -t + 2$ when t is in $[1, 2]$, and $f(t) = 0$ otherwise, then you write

$$f(t) = t\,\big(u(t) - u(t-1)\big) + (-t+2)\,\big(u(t-1) - u(t-2)\big).$$

How does the Heaviside function interact with the Laplace transform? We saw that (for $a \geq 0$)

$$\mathcal{L}\big\{u(t-a)\big\} = \frac{e^{-as}}{s}.$$

This computation can be generalized into a *shifting property* or *second shifting property*:

$$\boxed{\mathcal{L}\big\{f(t-a)\,u(t-a)\big\} = e^{-as}\mathcal{L}\big\{f(t)\big\} \qquad (a \geq 0).} \tag{6.1}$$

For example,

$$\mathcal{L}\big\{t\,u(t-1)\big\} = \mathcal{L}\big\{\big((t-1)+1\big)\,u(t-1)\big\} = e^{-s}\mathcal{L}\big\{t+1\big\} = e^{-s}\left(\frac{1}{s^2} + \frac{1}{s}\right).$$

Example 6.2.2: Consider the mass-spring system

$$x''(t) + x(t) = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

where $f(t) = 1$ if $1 \leq t < 5$ and zero otherwise. The function $f(t)$ is not periodic and is defined piecewise; no problem for Laplace. Imagine a rocket attached to the mass is fired for 4 seconds starting at $t = 1$. Or perhaps imagine an RLC circuit, where the voltage is raised at a constant rate for 4 seconds starting at $t = 1$ and then held steady again starting at $t = 5$ (recall that $f(t)$ represents the derivative of the voltage in the RLC circuit).

We write $f(t) = u(t-1) - u(t-5)$. We transform the equation and we plug in the initial conditions as before

$$s^2 X(s) + X(s) = \frac{e^{-s}}{s} - \frac{e^{-5s}}{s}.$$

We solve for $X(s)$,

$$X(s) = \frac{e^{-s}}{s(s^2+1)} - \frac{e^{-5s}}{s(s^2+1)}.$$

We leave it as an exercise to the reader to show that

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = 1 - \cos t.$$

In other words, $\mathcal{L}\{1-\cos t\} = \frac{1}{s(s^2+1)}$. Using the second shifting property (6.1), we find

$$\mathcal{L}^{-1}\left\{\frac{e^{-s}}{s(s^2+1)}\right\} = \mathcal{L}^{-1}\big\{e^{-s}\mathcal{L}\{1-\cos t\}\big\} = \big(1-\cos(t-1)\big)\,u(t-1).$$

Similarly,

$$\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{s(s^2+1)}\right\} = \mathcal{L}^{-1}\big\{e^{-5s}\mathcal{L}\{1-\cos t\}\big\} = \big(1-\cos(t-5)\big)\,u(t-5).$$

Hence, the solution is

$$x(t) = \big(1-\cos(t-1)\big)\,u(t-1) - \big(1-\cos(t-5)\big)\,u(t-5).$$

The plot of this solution is given in Figure 6.2.

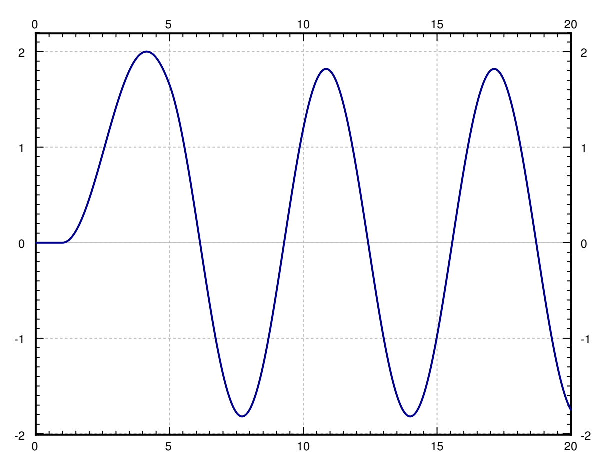

Figure 6.2: Plot of $x(t)$.

6.2.4 Transfer functions

The Laplace transform leads to the following useful concept for studying the steady-state behavior of a linear system. Consider an equation of the form

$$Lx = f(t),$$

where L is a linear constant-coefficient differential operator. Then $f(t)$ is usually thought of as the input of the system and $x(t)$ is thought of as the output of the system. For example, for a mass-spring system the input is the forcing function and the output is the behavior of the mass. We would like to have a convenient way to study the behavior of the system for different inputs.

Let us suppose that all the initial conditions are zero. We take the Laplace transform of the equation to obtain, for some $A(s)$,

$$A(s)X(s) = F(s).$$

Solving for the ratio ${X(s)}/{F(s)}$ gives the so-called *transfer function* $H(s) = {1}/{A(s)}$, that is,

$$H(s) = \frac{X(s)}{F(s)}.$$

In other words, $X(s) = H(s)F(s)$. We get an algebraic dependence of the output of the system based on the input. We can now easily study the steady-state behavior of the system given different inputs by simply multiplying by the transfer function. Moreover, it is possible to compute the $H(s)$ without knowing exactly what the equation is by observing the output $X(s)$ for a given input $F(s)$. Once $H(s)$ is known, you can find the output for any input.

Example 6.2.3: Given $x'' + \omega_0^2 x = f(t)$ (assume the initial conditions are zero), let us find the transfer function.

First, we take the Laplace transform of the equation,

$$s^2 X(s) + \omega_0^2 X(s) = F(s).$$

Now we solve for the transfer function ${X(s)}/{F(s)}$,

$$H(s) = \frac{X(s)}{F(s)} = \frac{1}{s^2 + \omega_0^2}.$$

Let us see how to use the transfer function. Suppose we have the constant input $f(t) = 1$. Hence $F(s) = {1}/{s}$, and

$$X(s) = H(s)F(s) = \frac{1}{s^2 + \omega_0^2} \frac{1}{s}.$$

Taking the inverse Laplace transform of $X(s)$, we obtain

$$x(t) = \frac{1 - \cos(\omega_0 t)}{\omega_0^2}.$$

Similarly, for any other input $F(s)$, the output is $X(s) = H(s)F(s) = \frac{1}{s^2+\omega_0^2} F(s)$.

6.2.5 Transforms of integrals

A feature of the Laplace transform is that it is able to deal with integral equations. These are equations in which integrals rather than derivatives of functions appear. The basic property, which can be proved by applying the definition and doing integration by parts, is

$$\boxed{\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}\,F(s).}$$

It is sometimes useful (e.g. for computing the inverse transform) to write this as

$$\int_0^t f(\tau)\,d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}\,F(s)\right\}.$$

Example 6.2.4: To compute $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\}$, we could apply this integration rule:

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\,\frac{1}{s^2+1}\right\} = \int_0^t \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\}\,d\tau = \int_0^t \sin\tau\,d\tau = 1-\cos t.$$

Example 6.2.5: An equation containing an integral of the unknown function is called an *integral equation*. Consider

$$x(t) - t = \int_0^t x(\tau)\,d\tau,$$

where we wish to solve for $x(t)$. We apply the Laplace transform to get

$$X(s) - \frac{1}{s^2} = \frac{1}{s}X(s),$$

where $X(s) = \mathcal{L}\{x(t)\}$. Thus

$$X(s) = \frac{1}{s(s-1)} = \frac{1}{s-1} - \frac{1}{s}.$$

The inverse Laplace transform gives

$$x(t) = e^t - 1.$$

6.2.6 Periodic functions

The reader might ask: What about periodic functions as our input $f(t)$? That is, a function $f(t)$ where $f(t) = f(t+P)$ for some constant P (the period). Well, let us compute $F(s)$:

$$\begin{aligned}
F(s) &= \int_0^\infty e^{-st} f(t)\,dt = \int_0^P e^{-st} f(t)\,dt + \int_P^\infty e^{-st} f(t)\,dt \\
&= \int_0^P e^{-st} f(t)\,dt + \int_0^\infty e^{-s(t+P)} f(t+P)\,dt = \int_0^P e^{-st} f(t)\,dt + e^{-Ps}\int_0^\infty e^{-st} f(t)\,dt \\
&= \int_0^P e^{-st} f(t)\,dt + e^{-Ps}F(s).
\end{aligned}$$

Solving for $F(s)$, we get

$$F(s) = \frac{1}{1-e^{-Ps}} \int_0^P e^{-st} f(t)\, dt.$$

As before, computing the inverse would be more complex and possibly involve consulting a table. Let us not worry about computing the inverse here.

Example 6.2.6: Suppose $f(t)$ is a version of the *sawtooth*, that is, let $f(t) = t$ for $0 \leq t < 1$ and use $f(t) = f(t+1)$ to extend it periodically. So $f(t) = t-1$ for $1 \leq t < 2$, $f(t) = t-2$ for $2 \leq t < 3$, etc. Then $P = 1$ and a short computation with integration by parts gets

$$F(s) = \frac{1}{1-e^{-s}} \int_0^1 e^{-st} t\, dt = \frac{1}{1-e^{-s}} \left(\frac{-e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} \right) = \frac{-e^{-s}}{(1-e^{-s})s} + \frac{1}{s^2}.$$

6.2.7 Exercises

***Exercise* 6.2.2:** *Using the Heaviside function write down the piecewise function that is 0 for $t < 0$, t^2 for t in $[0,1]$ and t for $t > 1$.*

***Exercise* 6.2.3:** *Using the Laplace transform solve*

$$mx'' + cx' + kx = 0, \quad x(0) = a, \quad x'(0) = b,$$

where $m > 0$, $c > 0$, $k > 0$, and $c^2 - 4km > 0$ (system is overdamped).

***Exercise* 6.2.4:** *Using the Laplace transform solve*

$$mx'' + cx' + kx = 0, \quad x(0) = a, \quad x'(0) = b,$$

where $m > 0$, $c > 0$, $k > 0$, and $c^2 - 4km < 0$ (system is underdamped).

***Exercise* 6.2.5:** *Using the Laplace transform solve*

$$mx'' + cx' + kx = 0, \quad x(0) = a, \quad x'(0) = b,$$

where $m > 0$, $c > 0$, $k > 0$, and $c^2 = 4km$ (system is critically damped).

***Exercise* 6.2.6:** *Solve $x'' + x = u(t-1)$ for initial conditions $x(0) = 0$ and $x'(0) = 0$.*

***Exercise* 6.2.7:** *Show the "differentiation of the transform" property. Suppose $\mathcal{L}\big\{f(t)\big\} = F(s)$, then show*

$$\mathcal{L}\big\{-t f(t)\big\} = F'(s).$$

Hint: Differentiate under the integral sign.

***Exercise* 6.2.8:** *Solve $x''' + x = t^3 u(t-1)$ for initial conditions $x(0) = 1$ and $x'(0) = 0$, $x''(0) = 0$.*

***Exercise* 6.2.9:** *Using the definition of the Laplace transform, justify the second shifting property: $\mathcal{L}\big\{f(t-a)\, u(t-a)\big\} = e^{-as} \mathcal{L}\big\{f(t)\big\}$.*

***Exercise* 6.2.10:** *Consider the mass-spring system with a rocket from Example 6.2.2. We noticed that the solution kept oscillating after the rocket stopped running. The amplitude of the oscillation depends on the time that the rocket was fired (for 4 seconds in the example).*

a) *Find a formula for the amplitude of the resulting oscillation in terms of the amount of time the rocket is fired.*

b) *Is there a nonzero time (if so, what is it?) for which the rocket fires and the resulting oscillation has amplitude 0 (the mass is not moving)?*

***Exercise* 6.2.11:** *Define*

$$f(t) = \begin{cases} (t-1)^2 & \text{if } 1 \le t < 2, \\ 3-t & \text{if } 2 \le t < 3, \\ 0 & \text{otherwise}. \end{cases}$$

a) *Sketch the graph of $f(t)$.*

b) *Write down $f(t)$ using the Heaviside function.*

c) *Solve $x'' + x = f(t)$, $x(0) = 0$, $x'(0) = 0$ using the Laplace transform.*

***Exercise* 6.2.12:** *Find the transfer function for $mx'' + cx' + kx = f(t)$, $x(0) = 0$, $x'(0) = 0$.*

***Exercise* 6.2.13:** *Suppose $Lx = f(t)$, $x(0) = 0$, $x'(0) = 0$ for the input function $f(t) = t$ gives the output $x(t) = 2e^{-t} + te^{-t} + (t-2)$.*

a) *Find $F(s)$, $X(s)$, and the transfer function $H(s)$.*

b) *If the input is instead $f(t) = \sin(t)$, find the new output $x(t)$.*

***Exercise* 6.2.14:** *Suppose $f(t) = 1$ if $0 \le t < 1$ and $f(t) = 0$ if $1 \le t < 2$, and then extend periodically for all $t \ge 0$ so that $f(t) = f(t+2)$. Compute the Laplace transform $F(s)$.*

***Exercise* 6.2.101:** *Using the Heaviside function $u(t)$, write down the function*

$$f(t) = \begin{cases} 0 & \text{if} \quad t < 1, \\ t-1 & \text{if } 1 \le t < 2, \\ 1 & \text{if } 2 \le t. \end{cases}$$

***Exercise* 6.2.102:** *Solve $x'' - x = (t^2 - 1)u(t-1)$ for initial conditions $x(0) = 1$, $x'(0) = 2$ using the Laplace transform.*

***Exercise* 6.2.103:** *Find the transfer function for $x' + x = f(t)$, $x(0) = 0$.*

***Exercise* 6.2.104:** *Suppose $Lx = f(t)$, $x(0) = 0$, $x'(0) = 0$ for the input function $f(t) = 4$ gives the output $x(t) = 1 - \cos(2t)$. Find $F(s)$, $X(s)$, and the transfer function $H(s)$.*

6.3 Convolution

Note: 1 or 1.5 lectures, §7.2 in [EP], §6.6 in [BD]

6.3.1 The convolution

The Laplace transformation of a product is not the product of the transforms. All hope is not lost, however. We simply have to use a different type of "product." Take two functions $f(t)$ and $g(t)$ defined for $t \geq 0$, and define the *convolution** of $f(t)$ and $g(t)$ as

$$\boxed{(f * g)(t) \overset{\text{def}}{=} \int_0^t f(\tau) g(t-\tau)\, d\tau.} \tag{6.2}$$

As you can see, the convolution of two functions of t is another function of t.

Example 6.3.1: Take $f(t) = e^t$ and $g(t) = t$ for $t \geq 0$. Then

$$(f * g)(t) = \int_0^t e^{\tau}(t-\tau)\, d\tau = e^t - t - 1.$$

To solve the integral we did one integration by parts.

Example 6.3.2: Take $f(t) = \sin(\omega t)$ and $g(t) = \cos(\omega t)$ for $t \geq 0$. Then

$$(f * g)(t) = \int_0^t \sin(\omega\tau)\, \cos\big(\omega(t-\tau)\big)\, d\tau.$$

Apply the identity

$$\cos(\theta)\sin(\psi) = \frac{1}{2}\big(\sin(\theta+\psi) - \sin(\theta-\psi)\big),$$

to get

$$\begin{aligned}
(f * g)(t) &= \int_0^t \frac{1}{2}\big(\sin(\omega t) - \sin(\omega t - 2\omega\tau)\big)\, d\tau \\
&= \left[\frac{1}{2}\tau\sin(\omega t) - \frac{1}{4\omega}\cos(\omega t - 2\omega\tau)\right]_{\tau=0}^{t} \\
&= \frac{1}{2}t\sin(\omega t).
\end{aligned}$$

The formula holds only for $t \geq 0$. The functions f, g, and $f * g$ are undefined for $t < 0$.

*For those who have seen convolution before, you may have seen it defined as $(f*g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\, d\tau$. This definition agrees with (6.2) if you define $f(t)$ and $g(t)$ to be zero for $t < 0$. When discussing the Laplace transform, the definition we gave is sufficient. Convolution does occur in many other applications, however, where you may have to use the more general definition with infinities.

Convolution has many properties that make it behave like a product. Let c be a constant and f, g, and h be functions. It is a calculus exercise to verify that

$$\begin{aligned}
&f * g = g * f,\\
&(cf) * g = f * (cg) = c(f * g),\\
&(f + g) * h = f * h + g * h,\\
&(f * g) * h = f * (g * h).
\end{aligned}$$

The first property means that $(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau = \int_0^t f(t-\tau)g(\tau)\,d\tau$. The most interesting property for us is the following theorem.

Theorem 6.3.1. *If $f(t)$ and $g(t)$ are of exponential order, then so is $(f * g)(t)$ and*

$$\boxed{\mathcal{L}\big\{(f * g)(t)\big\} = \mathcal{L}\left\{\int_0^t f(\tau)g(t-\tau)\,d\tau\right\} = \mathcal{L}\big\{f(t)\big\}\mathcal{L}\big\{g(t)\big\}.}$$

In other words, the Laplace transform of a convolution is the product of the Laplace transforms: $\mathcal{L}\big\{(f * g)(t)\big\} = F(s)G(s)$, or in reverse, $\mathcal{L}^{-1}\big\{F(s)G(s)\big\} = (f * g)(t)$.

Example 6.3.3: Suppose we wish to find the inverse Laplace transform of

$$\frac{1}{s^2(s+1)} = \frac{1}{s^2}\,\frac{1}{s+1}.$$

We recognize the two entries of Table 6.2:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t \qquad \text{and} \qquad \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}.$$

Therefore, we convolve t and e^{-t},

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\,\frac{1}{s+1}\right\} = \int_0^t \tau e^{-(t-\tau)}\,d\tau = e^{-t} + t - 1.$$

The calculation of the integral involved an integration by parts.

6.3.2 Solving ODEs

The next example demonstrates the full power of the convolution and the Laplace transform. We can give the solution to the forced oscillation problem for any forcing function as a definite integral.

Example 6.3.4: Find the solution to

$$x'' + \omega_0^2 x = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

for an arbitrary function $f(t)$.

We first apply the Laplace transform to the equation. Denote the transform of $x(t)$ by $X(s)$ and the transform of $f(t)$ by $F(s)$ as usual. We get

$$s^2X(s) + \omega_0^2 X(s) = F(s),$$

or in other words,

$$X(s) = \frac{1}{s^2 + \omega_0^2} F(s).$$

Recall that $H(s) = \frac{1}{s^2+\omega_0^2}$ is the transfer function. We know

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + \omega_0^2}\right\} = \frac{\sin(\omega_0 t)}{\omega_0}.$$

Therefore,

$$x(t) = \int_0^t \frac{\sin(\omega_0 \tau)}{\omega_0} f(t-\tau)\, d\tau,$$

or, if we reverse the order,

$$x(t) = \int_0^t f(\tau) \frac{\sin\bigl(\omega_0 (t-\tau)\bigr)}{\omega_0}\, d\tau.$$

Notice one more feature of the example above. We can now see how the Laplace transform handles resonance. Suppose that $f(t) = \cos(\omega_0 t)$. Then

$$x(t) = \int_0^t \frac{\sin(\omega_0 \tau)}{\omega_0} \cos\bigl(\omega_0 (t-\tau)\bigr)\, d\tau = \frac{1}{\omega_0} \int_0^t \sin(\omega_0 \tau) \cos\bigl(\omega_0 (t-\tau)\bigr)\, d\tau.$$

We have computed the convolution of sine and cosine in Example 6.3.2. Hence

$$x(t) = \left(\frac{1}{\omega_0}\right) \left(\frac{1}{2}\, t \sin(\omega_0 t)\right) = \frac{1}{2\omega_0}\, t \sin(\omega_0 t).$$

Note the t in front of the sine. The solution, therefore, grows without bound as t gets large, meaning we get resonance.

The general idea here is that if $H(s)$ is the transfer function, then $X(s) = H(s)F(s)$. If we find the $h(t) = \mathcal{L}^{-1}\bigl\{H(s)\bigr\}$, then

$$x(t) = \mathcal{L}^{-1}\bigl\{X(s)\bigr\} = \mathcal{L}^{-1}\bigl\{F(s)H(s)\bigr\} = (f * h)(t) = \int_0^t f(\tau)h(t-\tau)\, d\tau.$$

Hence, we can solve any constant-coefficient equation with an arbitrary forcing function $f(t)$ as a definite integral using convolution. A definite integral, rather than a closed-form solution, is usually enough for most practical purposes. It is not hard to numerically evaluate a definite integral.

6.3.3 Volterra integral equation

An integral equation involving convolution that comes up, for example, when studying systems with memory effects is the *Volterra integral equation*[*]

$$x(t) = f(t) + \int_0^t g(t-\tau)x(\tau)\, d\tau.$$

Here $f(t)$ and $g(t)$ are known functions and $x(t)$ is an unknown we wish to solve for. We have $x(t) = f(t) + (g * x)(t)$, so the question is: If convolution is like a product, can we also "divide" to solve this equation? We apply the Laplace transform to the equation to obtain

$$X(s) = F(s) + G(s)X(s),$$

where $X(s)$, $F(s)$, and $G(s)$ are the Laplace transforms of $x(t)$, $f(t)$, and $g(t)$, respectively. We find

$$X(s) = \frac{F(s)}{1 - G(s)}.$$

To find $x(t)$, we now need to find the inverse Laplace transform of $X(s)$.

Example 6.3.5: Solve

$$x(t) = e^{-t} + \int_0^t \sinh(t-\tau)x(\tau)\, d\tau.$$

We apply the Laplace transform to obtain

$$X(s) = \frac{1}{s+1} + \frac{1}{s^2-1}X(s),$$

or

$$X(s) = \frac{\frac{1}{s+1}}{1 - \frac{1}{s^2-1}} = \frac{s-1}{s^2-2} = \frac{s}{s^2-2} - \frac{1}{s^2-2}.$$

It is not hard to apply Table 6.1 on page 295 to find

$$x(t) = \cosh\bigl(\sqrt{2}\, t\bigr) - \frac{1}{\sqrt{2}} \sinh\bigl(\sqrt{2}\, t\bigr).$$

6.3.4 Exercises

***Exercise* 6.3.1:** *Let $f(t) = t^2$ for $t \geq 0$, and $g(t) = u(t-1)$. Compute $f * g$.*

***Exercise* 6.3.2:** *Let $f(t) = t$ for $t \geq 0$, and $g(t) = \sin t$ for $t \geq 0$. Compute $f * g$.*

***Exercise* 6.3.3:** *Find the solution to*

$$mx'' + cx' + kx = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

for an arbitrary function $f(t)$, where $m > 0$, $c > 0$, $k > 0$, and $c^2 - 4km > 0$ (the system is overdamped). Write the solution as a definite integral.

[*]Named for the Italian mathematician Vito Volterra (1860–1940).

***Exercise* 6.3.4:** *Find the solution to*

$$mx'' + cx' + kx = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

for an arbitrary function $f(t)$, where $m > 0$, $c > 0$, $k > 0$, and $c^2 - 4km < 0$ (the system is underdamped). Write the solution as a definite integral.

***Exercise* 6.3.5:** *Find the solution to*

$$mx'' + cx' + kx = f(t), \quad x(0) = 0, \quad x'(0) = 0,$$

for an arbitrary function $f(t)$, where $m > 0$, $c > 0$, $k > 0$, and $c^2 = 4km$ (the system is critically damped). Write the solution as a definite integral.

***Exercise* 6.3.6:** *Solve*

$$x(t) = e^{-t} + \int_0^t \cos(t-\tau)x(\tau)\, d\tau.$$

***Exercise* 6.3.7:** *Solve*

$$x(t) = \cos t + \int_0^t \cos(t-\tau)x(\tau)\, d\tau.$$

***Exercise* 6.3.8:** *Compute $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+4)^2}\right\}$ using convolution.*

***Exercise* 6.3.9:** *Write down the solution to $x'' - 2x = e^{-t^2}$, $x(0) = 0$, $x'(0) = 0$ as a definite integral. Hint: Do not try to compute the Laplace transform of e^{-t^2}.*

***Exercise* 6.3.101:** *Let $f(t) = \cos t$ for $t \geq 0$, and $g(t) = e^{-t}$. Compute $f * g$.*

***Exercise* 6.3.102:** *Compute $\mathcal{L}^{-1}\left\{\frac{5}{s^4+s^2}\right\}$ using convolution.*

***Exercise* 6.3.103:** *Solve $x'' + x = \sin t$, $x(0) = 0$, $x'(0) = 0$ using convolution.*

***Exercise* 6.3.104:** *Solve $x''' + x' = f(t)$, $x(0) = 0$, $x'(0) = 0$, $x''(0) = 0$ using convolution. Write the result as a definite integral.*

6.4 Dirac delta and impulse response

Note: 1 or 1.5 lecture, §7.6 in [EP], §6.5 in [BD]

6.4.1 Rectangular pulse

Often we study a physical system by putting in a short pulse and then seeing what the system does. The resulting behavior is often called *impulse response*, and understanding it tells us how the system responds to any input. Let us see what we mean by a pulse. The simplest kind of a pulse is a *rectangular pulse* defined by

$$\varphi(t) = \begin{cases} 0 & \text{if } \quad t < a, \\ M & \text{if } \ a \leq t < b, \\ 0 & \text{if } \ b \leq t. \end{cases}$$

See Figure 6.3 for a graph.

Notice that

$$\varphi(t) = M\big(u(t-a) - u(t-b)\big),$$

where $u(t)$ is the unit step function. The Laplace transform of a rectangular pulse is

$$\mathcal{L}\big\{\varphi(t)\big\} = \mathcal{L}\big\{M\big(u(t-a) - u(t-b)\big)\big\} = M\frac{e^{-as} - e^{-bs}}{s}.$$

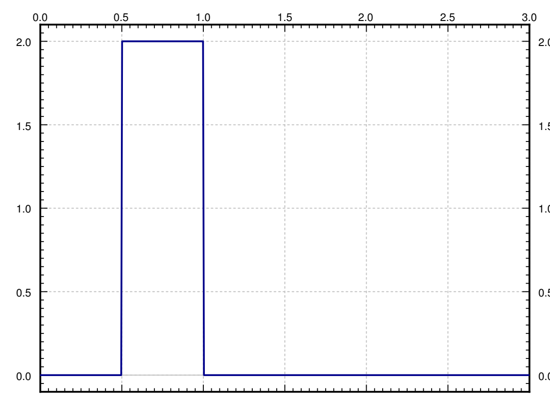

Figure 6.3: Sample rectangular pulse with $a = 0.5$, $b = 1$, and $M = 2$.

For simplicity, let $a = 0$. We also set $M = 1/b$ so that

$$\int_0^\infty \varphi(t)\,dt = 1.$$

That is, we have a pulse of "unit mass." This way for any b the pulse has the same "oomph." For such a pulse,

$$\mathcal{L}\big\{\varphi(t)\big\} = \mathcal{L}\left\{\frac{u(t) - u(t-b)}{b}\right\} = \frac{1 - e^{-bs}}{bs}.$$

We want b to be very small; we wish to have the pulse be very short and very tall. By letting b go to zero, we arrive at the concept of the Dirac delta function. The limit of $\frac{1-e^{-bs}}{bs}$ as $b \to 0$ is 1, so we are looking for a "function" whose Laplace transform is 1.

6.4.2 The delta function

The *Dirac delta function*[*] is not exactly a function; it is sometimes called a *generalized function*. We avoid unnecessary details and simply say that it is an object that does not really make sense unless we integrate it. The motivation is that we would like a "function" $\delta(t)$ such that for any continuous function $f(t)$,

$$\boxed{\int_{-\infty}^{\infty} \delta(t) f(t)\, dt = f(0).}$$

The formula should hold if we integrate over any interval that contains 0, not just $(-\infty, \infty)$. So $\delta(t)$ is a "function" with all its "mass" at the single point $t = 0$. For any interval[†] $[c, d]$,

$$\int_{c}^{d} \delta(t)\, dt = \begin{cases} 1 & \text{if the interval } [c,d] \text{ contains 0, i.e. } c \leq 0 \leq d, \\ 0 & \text{otherwise.} \end{cases}$$

Unfortunately, there is no such function in the classical sense. You could informally think that $\delta(t)$ is zero for $t \neq 0$ and somehow infinite at $t = 0$.

A good way to think about $\delta(t)$ is as a limit of pulses of decreasing length, each having an integral of 1. For example, consider a rectangular pulse $\varphi(t)$ as above with $a = 0$ and $M = 1/b$, that is, $\varphi(t) = \frac{u(t)-u(t-b)}{b}$. Compute

$$\int_{-\infty}^{\infty} \varphi(t) f(t)\, dt = \int_{-\infty}^{\infty} \frac{u(t)-u(t-b)}{b} f(t)\, dt = \frac{1}{b}\int_{0}^{b} f(t)\, dt.$$

If $f(t)$ is continuous at $t = 0$, then for very small b, the function $f(t)$ is approximately equal to $f(0)$ on the interval $[0, b]$. We approximate the integral

$$\frac{1}{b}\int_{0}^{b} f(t)\, dt \approx \frac{1}{b}\int_{0}^{b} f(0)\, dt = f(0).$$

Hence,

$$\lim_{b\to 0} \int_{-\infty}^{\infty} \varphi(t) f(t)\, dt = \lim_{b\to 0} \frac{1}{b}\int_{0}^{b} f(t)\, dt = f(0).$$

Let us therefore accept $\delta(t)$ as an object that is possible to integrate. We often want to shift δ to another point, for example $\delta(t-a)$. In that case,

$$\int_{-\infty}^{\infty} \delta(t-a) f(t)\, dt = f(a).$$

Note that $\delta(a-t)$ is the same object as $\delta(t-a)$. With that, the convolution of $\delta(t)$ with $f(t)$ is again $f(t)$,

$$(f * \delta)(t) = \int_{0}^{t} f(\tau)\delta(t-\tau)\, d\tau = f(t).$$

[*] Named after the English physicist and mathematician Paul Adrien Maurice Dirac (1902–1984).

[†] It is important that we consider c and d as part of the interval. One could write this integral as $\int_{c-}^{d+}$.

As we can integrate $\delta(t)$, we compute its Laplace transform:

$$\boxed{\mathcal{L}\{\delta(t-a)\} = \int_0^\infty e^{-st}\delta(t-a)\,dt = e^{-as}.}$$

In particular,

$$\mathcal{L}\{\delta(t)\} = 1.$$

Remark 6.4.1: The Laplace transform of $\delta(t-a)$ would be the Laplace transform of the derivative of the Heaviside function $u(t-a)$, if the Heaviside function had a derivative. First,

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}.$$

To obtain what the Laplace transform of the derivative would be, we multiply by s to obtain e^{-as}, which is the Laplace transform of $\delta(t-a)$. We see the same thing using integration,

$$\int_0^t \delta(s-a)\,ds = u(t-a).$$

So in a certain sense

$$\text{``}\quad \frac{d}{dt}\Big[u(t-a)\Big] = \delta(t-a). \quad\text{''}$$

This line of reasoning allows us to talk about derivatives of functions with jump discontinuities. We can think of the derivative of the Heaviside function $u(t-a)$ as being somehow infinite at a, which is precisely our intuitive understanding of the delta function.

Example 6.4.1: Let us compute $\mathcal{L}^{-1}\left\{\frac{s+1}{s}\right\}$. So far, we only computed the inverse transform of proper rational functions in the s variable. That is, the numerator was of lower degree than the denominator. Not so with $\frac{s+1}{s}$. We can use the delta function to compute,

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s}\right\} = \mathcal{L}^{-1}\left\{1+\frac{1}{s}\right\} = \mathcal{L}^{-1}\{1\} + \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = \delta(t) + 1.$$

The resulting object is a generalized function and only makes sense when put underneath an integral.

6.4.3 Impulse response

As we said before, in the differential equation $Lx = f(t)$, we think of $f(t)$ as the input, and $x(t)$ as the output. We think of the delta function as an impulse, and so to find the response to an impulse, we use the delta function in place of $f(t)$. The solution to

$$Lx = \delta(t)$$

is called the *impulse response*.

Example 6.4.2: Solve (find the impulse response)

$$x'' + \omega_0^2 x = \delta(t), \quad x(0) = 0, \quad x'(0) = 0. \tag{6.3}$$

Apply the Laplace transform to the equation, and denote the transform of $x(t)$ by $X(s)$:

$$s^2 X(s) + \omega_0^2 X(s) = 1, \qquad \text{and so} \qquad X(s) = \frac{1}{s^2 + \omega_0^2}.$$

The inverse Laplace transform produces (for $t > 0$)

$$x(t) = \frac{\sin(\omega_0 t)}{\omega_0}.$$

Remark 6.4.2: Perhaps an astute reader will notice that it does not seem like $x'(0) = 0$. However, we really want to think that $x(t) = 0$ for $t \leq 0$, so $x(t)$ has a "corner" at $t = 0$, that is, x' has a jump discontinuity there, which is what produces the $\delta(t)$ when we take x''. See Remark 6.4.1. The initial condition really is $x'(0-) = \lim_{t \uparrow 0} x'(0) = 0$.

Let us notice something about the example above. In Example 6.3.4, we found that when the input is $f(t)$, the solution to $Lx = f(t)$ is given by

$$x(t) = \int_0^t f(\tau) \frac{\sin\bigl(\omega_0 (t - \tau)\bigr)}{\omega_0} \, d\tau.$$

That is, *the solution for an arbitrary input is given as convolution with the impulse response*. Let us see why. For functions $f(t)$ and $h(t)$ that are zero for negative t,

$$(f * h)''(t) = \frac{d^2}{dt^2} \left[\int_0^t f(\tau) h(t - \tau) \, d\tau \right] = \int_0^t f(\tau) h''(t - \tau) \, d\tau = (f * h'')(t).$$

We differentiate twice under the integral*. Suppose that $h(t)$ is the impulse response (solution to $h'' + \omega_0^2 h = \delta(t)$). If we convolve the entire equation (6.3), the left-hand side becomes

$$f * (h'' + \omega_0^2 h) = (f * h'') + \omega_0^2 (f * h) = (f * h)'' + \omega_0^2 (f * h).$$

The right-hand side becomes

$$(f * \delta)(t) = f(t).$$

Therefore, if h is the impulse response, then $x(t) = (f * h)(t)$ is the solution to

$$x'' + \omega_0^2 x = f(t).$$

The procedure works for any constant-coefficient linear equation $Lx = f(t)$. If you find the impulse response h (solution to $Lh = \delta(t)$), you also know how to obtain the output $x(t)$ for any input $f(t)$. We simply convolve, $x(t) = (f * h)(t)$. As you may have noticed in the example, the impulse response is in fact just the inverse Laplace transform of the transfer function, that is, $h(t) = \mathcal{L}^{-1}\bigl\{H(s)\bigr\}$.

---

*You may be worried about the fact that our h is not really twice differentiable, in fact, $h'(0)$ does not exist as a normal derivative, and h'' involves the δ function. You can pretend that h is twice differentiable for this computation and either apply the standard Leibniz rule (that $h(t) = 0$ for $t < 0$ is important) or note that for such functions, the integral is actually over $(-\infty, \infty)$ rather than over $[0, t]$.

6.4.4 Three-point beam bending

A quite different example application where the delta function appears is in representing point loads on a steel beam. Consider a beam of length L, resting on two simple supports at the ends. Let x denote the position on the beam, and let $y(x)$ denote the deflection of the beam in the vertical direction. The deflection $y(x)$ satisfies the *Euler–Bernoulli equation**,

$$EI\frac{d^4y}{dx^4} = F(x),$$

where E and I are constants† and $F(x)$ is the force applied per unit length at position x. The situation we are interested in is when the force is applied at a single point as in Figure 6.4.

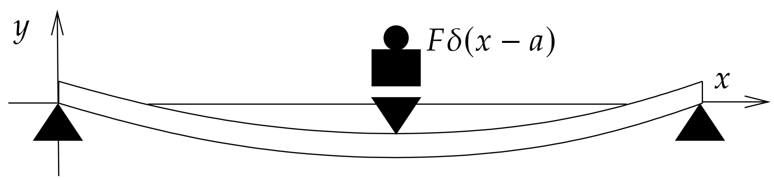

Figure 6.4: Three-point bending.

The equation becomes

$$EI\frac{d^4y}{dx^4} = -F\delta(x-a),$$

where $x = a$ is the point where the mass is applied. The constant F is the force applied and the minus sign indicates that the force is downward, that is, in the negative y direction. The end points of the beam satisfy the conditions,

$$y(0) = 0, \qquad y''(0) = 0,$$
$$y(L) = 0, \qquad y''(L) = 0.$$

See § 5.2 for further information about endpoint conditions applied to beams.

Example 6.4.3: Suppose that the length of the beam is 2, and $EI = 1$ for simplicity. Further suppose that a force $F = 1$ is applied at $x = 1$. That is, we have the equation

$$\frac{d^4y}{dx^4} = -\delta(x-1),$$

and the endpoint conditions are

$$y(0) = 0, \qquad y''(0) = 0, \qquad y(2) = 0, \qquad y''(2) = 0.$$

*Named for the Swiss mathematicians Jacob Bernoulli (1654–1705), Daniel Bernoulli (1700–1782), the nephew of Jacob, and Leonhard Paul Euler (1707–1783).

†E is the elastic modulus and I is the second moment of area. Let us not worry about the details and simply think of these as some given constants.

We could integrate, but using the Laplace transform is even easier. We apply the transform in the x variable rather than the t variable. We again denote the transform of $y(x)$ by $Y(s)$.

$$s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0) = -e^{-s}.$$

We notice that $y(0) = 0$ and $y''(0) = 0$. Let us call $C_1 = y'(0)$ and $C_2 = y'''(0)$. We solve for $Y(s)$,

$$Y(s) = \frac{-e^{-s}}{s^4} + \frac{C_1}{s^2} + \frac{C_2}{s^4}.$$

We take the inverse Laplace transform utilizing the second shifting property (6.1) for the first term:

$$y(x) = \frac{-(x-1)^3}{6} u(x-1) + C_1 x + \frac{C_2}{6} x^3.$$

We still need to apply two of the endpoint conditions. As the conditions are at $x = 2$ we can simply replace $u(x-1) = 1$ when taking the derivatives. Therefore,

$$0 = y(2) = \frac{-(2-1)^3}{6} + C_1(2) + \frac{C_2}{6} 2^3 = \frac{-1}{6} + 2C_1 + \frac{4}{3} C_2,$$

and

$$0 = y''(2) = \frac{-3 \cdot 2 \cdot (2-1)}{6} + \frac{C_2}{6} 3 \cdot 2 \cdot 2 = -1 + 2C_2.$$

Hence, $C_2 = \frac{1}{2}$. Solving for C_1 using the first equation, we obtain $C_1 = \frac{-1}{4}$. Our solution for the beam deflection is

$$y(x) = \frac{-(x-1)^3}{6} u(x-1) - \frac{x}{4} + \frac{x^3}{12}.$$

6.4.5 Exercises

***Exercise* 6.4.1:** *Solve (find the impulse response)* $x'' + x' + x = \delta(t)$, $x(0) = 0$, $x'(0) = 0$.

***Exercise* 6.4.2:** *Solve (find the impulse response)* $x'' + 2x' + x = \delta(t)$, $x(0) = 0$, $x'(0) = 0$.

***Exercise* 6.4.3:** *A pulse can come later and can be bigger. Solve* $x'' + 4x = 4\delta(t-1)$, $x(0) = 0$, $x'(0) = 0$.

***Exercise* 6.4.4:** *Suppose that* $f(t)$ *and* $g(t)$ *are differentiable functions (and the derivatives are continuous) and suppose that* $f(t) = g(t) = 0$ *for all* $t \leq 0$. *Show that*

$$(f * g)'(t) = (f' * g)(t) = (f * g')(t).$$

***Exercise* 6.4.5:** *Suppose that* $Lx = \delta(t)$, $x(0) = 0$, $x'(0) = 0$, *has the solution* $x(t) = te^{-t}$ *for* $t > 0$. *Find the solution to* $Lx = t^2$, $x(0) = 0$, $x'(0) = 0$ *for* $t > 0$.

***Exercise* 6.4.6:** *Compute* $\mathcal{L}^{-1}\left\{\frac{s^2+s+1}{s^2}\right\}$.

Exercise* 6.4.7** (challenging):*** *Solve Example 6.4.3 via integrating 4 times in the x variable.*

Exercise* 6.4.8*: *Suppose we have a beam of length 1 simply supported at the ends and suppose that a force $F = 1$ is applied at $x = 3/4$ in the downward direction. Suppose that $EI = 1$ for simplicity. Find the beam deflection $y(x)$.*

Exercise* 6.4.101*: *Solve (find the impulse response) $x'' = \delta(t)$, $x(0) = 0$, $x'(0) = 0$.*

Exercise* 6.4.102*: *Solve (find the impulse response) $x' + ax = \delta(t)$, $x(0) = 0$.*

Exercise* 6.4.103*: *Suppose that $Lx = \delta(t)$, $x(0) = 0$, $x'(0) = 0$, has the solution $x(t) = e^t \sin(t)$ for $t > 0$. Find (in closed form) the solution to $Lx = e^t$, $x(0) = 0$, $x'(0) = 0$ for $t > 0$.*

Exercise* 6.4.104*: *Compute $\mathcal{L}^{-1}\left\{\frac{s^2}{s^2+1}\right\}$.*

Exercise* 6.4.105*: *Compute $\mathcal{L}^{-1}\left\{\frac{3s^2e^{-s}+2}{s^2}\right\}$.*

6.5 Solving PDEs with the Laplace transform

Note: 1–1.5 lecture, can be skipped

The Laplace transform comes from the same family of transforms as does the Fourier series*, which we used in chapter 4 to solve partial differential equations (PDEs). It is therefore not surprising that we can also solve PDEs with the Laplace transform.

Given a PDE in two independent variables x and t, we use the Laplace transform on one of the variables (taking the transform of everything in sight), and derivatives in that variable become multiplications by the transformed variable s. The PDE becomes an ODE, which we solve. Afterwards, we invert the transform to find a solution to the original problem. It is best to see the procedure in an example.

Example 6.5.1: Consider the first-order PDE

$$y_t = -\alpha y_x, \qquad \text{for } x > 0, \;\; t > 0,$$

with side conditions

$$y(0,t) = C, \qquad y(x,0) = 0.$$

We will assume $\alpha > 0$ is a constant. This equation is called the *convection equation* or sometimes the *transport equation*, and it already made an appearance in § 1.9, with different conditions. See Figure 6.5 for a diagram of the setup.

A physical setup of this equation is a river of viscous goo, as we do not want anything to diffuse. The function y is the concentration of some toxic substance†. The variable x denotes position, where $x = 0$ is the location of a factory spewing the toxic substance into the river. The toxic substance flows into the river so that at $x = 0$ the concentration is always C. We wish to see what happens past the factory, that is, at $x > 0$. Let t be the time, and assume the factory started operations at $t = 0$, so that at $t = 0$ the river is just pure goo.

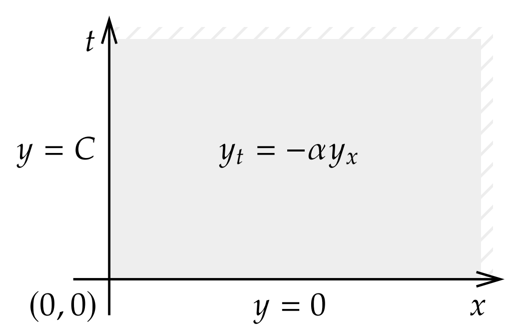

Figure 6.5: Transport equation on a half line.

Consider a function of two variables $y(x,t)$. Let us fix x and transform the t variable. For convenience, we treat the transformed s variable as a parameter, since there are no derivatives in s. That is, we write $Y(x)$ for the transformed function, and treat it as a function of x, leaving s as a parameter.

$$Y(x) = \mathcal{L}\big\{y(x,t)\big\} = \int_0^\infty y(x,t)e^{-st}\,dt.$$

*There is also a Fourier transform on the real line that looks sort of like the Laplace transform.
†It's a river of goo already. We're not hurting the environment much more.

The transform of a derivative with respect to x is just the derivative of the transformed function:

$$\mathcal{L}\big\{y_x(x,t)\big\} = \int_0^\infty y_x(x,t)e^{-st}\,dt = \frac{d}{dx}\left[\int_0^\infty y(x,t)e^{-st}\,dt\right] = Y'(x).$$

To transform the derivative in t (the variable being transformed), we use the rules from § 6.2:

$$\mathcal{L}\big\{y_t(x,t)\big\} = sY(x) - y(x,0).$$

In our specific case, $y(x,0) = 0$, and so $\mathcal{L}\big\{y_t(x,t)\big\} = sY(x)$. We transform the equation to find

$$sY(x) = -\alpha Y'(x).$$

This ODE needs an initial condition. The initial condition is the other side condition of the PDE, the one that depends on x. Everything is transformed, so we must also transform this condition

$$Y(0) = \mathcal{L}\big\{y(0,t)\big\} = \mathcal{L}\big\{C\big\} = \frac{C}{s}.$$

We solve the ODE problem $sY(x) = -\alpha Y'(x)$, $Y(0) = \frac{C}{s}$, to find

$$Y(x) = \frac{C}{s}e^{-\frac{s}{\alpha}x}.$$

We are not done. We have $Y(x)$, but we really want $y(x,t)$. We transform the s variable back to t. Let

$$u(t) = \begin{cases} 0 & \text{if } t < 0, \\ 1 & \text{otherwise} \end{cases}$$

be the Heaviside function. As

$$\mathcal{L}\big\{u(t-a)\big\} = \int_0^\infty u(t-a)\,e^{-st}\,dt = \int_a^\infty e^{-st}\,dt = \frac{e^{-as}}{s},$$

then

$$y(x,t) = \mathcal{L}^{-1}\left\{\frac{C}{s}e^{-\frac{s}{\alpha}x}\right\} = Cu\big(t - {}^{x}/_{\alpha}\big).$$

In other words,

$$y(x,t) = \begin{cases} 0 & \text{if } t < {}^{x}/_{\alpha}, \\ C & \text{otherwise.} \end{cases}$$

See Figure 6.6 on the next page for a diagram of this solution. The line of slope ${}^{1}/_{\alpha}$ indicates the wavefront of the toxic substance in the picture as it is leaving the factory. What the equation does is to try to move the initial condition to the right at speed α, and so it moves the boundary condition (the wavefront) too.

Shhh. . . y is not differentiable. It is not even continuous (nobody ever seems to notice). How could we plug something that's not differentiable into the equation? Well, just think

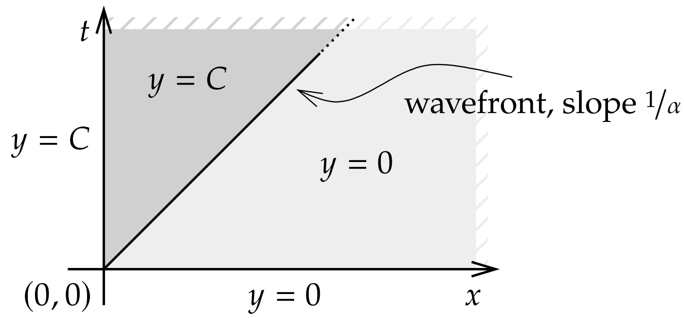

Figure 6.6: Wavefront of toxic substance is a line of slope $1/\alpha$.

of a differentiable function very, very close to y. Or, if you recognize the derivative of the Heaviside function as the delta function, then all is well too:

$$y_t(x,t) = \frac{\partial}{\partial t}\left[Cu\big(t - x/\alpha\big)\right] = Cu'\big(t - x/\alpha\big) = C\delta\big(t - x/\alpha\big)$$

and

$$y_x(x,t) = \frac{\partial}{\partial x}\left[Cu\big(t - x/\alpha\big)\right] = -\frac{C}{\alpha}u'\big(t - x/\alpha\big) = -\frac{C}{\alpha}\delta\big(t - x/\alpha\big).$$

So $y_t = -\alpha y_x$.

The Laplace transform is very good with constant-coefficient equations. One advantage of Laplace is that it easily handles nonhomogeneous side conditions. Let us try a more complicated example.

Example 6.5.2: Consider

$$\begin{aligned} y_t + y_x + y &= 0, &\qquad &\text{for } x > 0, \ \ t > 0, \\ y(0,t) &= \sin(t), & &y(x,0) = 0. \end{aligned}$$

Again, we transform t, and we write $Y(x)$ for the transformed function. As $y(x,0) = 0$, we find

$$sY(x) + Y'(x) + Y(x) = 0, \qquad Y(0) = \frac{1}{s^2+1}.$$

The solution of the transformed equation is

$$Y(x) = \frac{1}{s^2+1}e^{-(s+1)x} = \frac{1}{s^2+1}e^{-xs}e^{-x}.$$

Using the second shifting property (6.1) and linearity of the transform, we obtain the solution

$$y(x,t) = e^{-x}\sin(t-x)u(t-x).$$

We can also detect when the problem is *ill-posed* in the sense that it has no solution. Let us change the equation to

$$
\begin{aligned}
-y_t + y_x &= 0, \qquad \text{for } x > 0, \;\; t > 0, \\
y(0,t) &= \sin(t), \qquad y(x,0) = 0.
\end{aligned}
$$

Then the problem has no solution. First, let us see this in the language of § 1.9. The characteristic curves are $t = -x + C$. If τ is the characteristic coordinate, then we find the equation $y_\tau = 0$ along the curve, meaning a solution is constant along characteristic curves. But these curves intersect both the x-axis and the t-axis. For example, the curve $t = -x + 1$ intersects at $(1,0)$ and $(0,1)$. The solution is constant along the curve so $y(1,0)$ should equal $y(0,1)$. But $y(1,0) = 0$ and $y(0,1) = \sin(1) \neq 0$. See Figure 6.7.

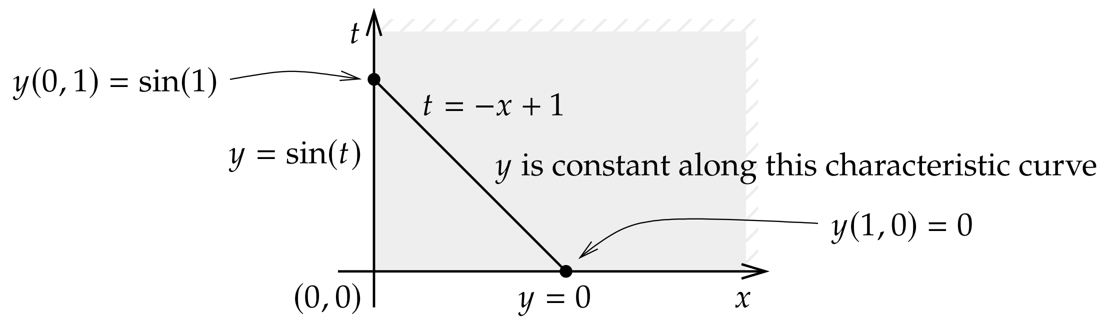

Figure 6.7: Ill-posed problem.

Now consider the transform. The transformed problem is

$$-sY(x) + Y'(x) = 0, \quad Y(0) = \frac{1}{s^2+1},$$

and the solution ought to be

$$Y(x) = \frac{1}{s^2+1} e^{sx}.$$

Importantly, this Laplace transform does not decay to zero at infinity! That is, since $x > 0$ in the region of interest,

$$\lim_{s\to\infty} \frac{1}{s^2+1} e^{sx} = \infty \neq 0.$$

It almost looks as if we could use the shifting property, but notice that the shift is in the wrong direction.

Of course, we need not restrict ourselves to first-order equations, although the computations become more involved for higher-order equations.

Example 6.5.3: Let us use Laplace for the following problem:

$$\begin{aligned} &y_t = y_{xx}, \qquad 0 < x < \infty, \;\; t > 0, \\ &y_x(0,t) = f(t), \\ &y(x,0) = 0. \end{aligned}$$

This problem corresponds to a half-infinite insulated rod with a given heat flux at one end. The setup would come up in real life in the case of a sufficiently long rod, where we would not consider a long enough time for our solution to notice that we do not have an infinite rod but just a very long one. In any case, the real-life situation imposes other conditions on our solution y. For example, we will assume that the solution is bounded. This boundedness condition will stand in for a boundary condition at the infinite end. Boundedness is also sufficient for the Laplace transform in the t variable to exist.

Transform the equation in the t variable to find

$$sY(x) = Y''(x).$$

The general solution to this ODE is

$$Y(x) = Ae^{\sqrt{s}x} + Be^{-\sqrt{s}x}.$$

Note that A and B depend on s. As y is bounded, then $Y(x)$ is bounded for any fixed $s > 0$, so for $Y(x)$ to stay bounded as $x \to \infty$, we must have $A = 0$.

Now consider the boundary condition at $x = 0$. Transform $Y'(0) = F(s)$ and so $-\sqrt{s}B = F(s)$. In other words,

$$Y(x) = F(s)\frac{-1}{\sqrt{s}}e^{-\sqrt{s}x}.$$

If we look up the inverse transform in a table such as the one in Appendix B (or we spend the afternoon doing calculus), we find

$$\mathcal{L}^{-1}\left[\frac{1}{\sqrt{s}}e^{-\sqrt{s}x}\right] = \frac{1}{\sqrt{\pi t}}e^{\frac{-x^2}{4t}}.$$

So

$$y(x,t) = \mathcal{L}^{-1}\left[F(s)\frac{-1}{\sqrt{s}}e^{-\sqrt{s}x}\right] = \int_0^t f(\tau)\frac{-1}{\sqrt{\pi(t-\tau)}}e^{\frac{-x^2}{4(t-\tau)}}\,d\tau.$$

Laplace can solve problems where separation of variables fails. Laplace does not mind nonhomogeneity, but it is essentially only useful for constant-coefficient equations.

6.5.1 Exercises

***Exercise* 6.5.1:** *Solve*

$$\begin{aligned} &y_t + y_x = 1, \qquad 0 < x < \infty, \;\; t > 0, \\ &y(0,t) = 1, \quad y(x,0) = 0. \end{aligned}$$

***Exercise* 6.5.2:** *Solve*

$$\begin{aligned} &y_t + \alpha y_x = 0, \qquad 0 < x < \infty, \ \ t > 0, \\ &y(0,t) = t, \quad y(x,0) = 0. \end{aligned}$$

***Exercise* 6.5.3:** *Solve*

$$\begin{aligned} &y_t + 2y_x = x + t, \qquad 0 < x < \infty, \ \ t > 0, \\ &y(0,t) = 0, \quad y(x,0) = 0. \end{aligned}$$

***Exercise* 6.5.4:** *For an $\alpha > 0$, solve*

$$\begin{aligned} &y_t + \alpha y_x + y = 0, \qquad 0 < x < \infty, \ \ t > 0, \\ &y(0,t) = \sin(t), \quad y(x,0) = 0. \end{aligned}$$

***Exercise* 6.5.5:** *Find the corresponding ODE problem for $Y(x)$, after transforming the t variable*

$$\begin{aligned} &y_{tt} + 3y_{xx} + y_{xt} + 3y_x + y = \sin(x) + t, \qquad 0 < x < 1, \ \ t > 0, \\ &y(0,t) = 1, \quad y(1,t) = t, \quad y(x,0) = 1 - x, \quad y_t(x,0) = 1. \end{aligned}$$

Do not solve the problem.

***Exercise* 6.5.6:** *Write down a solution to*

$$\begin{aligned} &y_t = y_{xx}, \qquad 0 < x < \infty, \ \ t > 0, \\ &y_x(0,t) = e^{-t}, \quad y(x,0) = 0, \end{aligned}$$

as a definite integral (convolution). Assume that y is bounded.

***Exercise* 6.5.7:** *Use the Laplace transform in t to solve*

$$\begin{aligned} &y_{tt} = y_{xx}, \qquad -\infty < x < \infty, \ \ t > 0, \\ &y(x,0) = 0, \quad y_t(x,0) = \sin(x). \end{aligned}$$

Assume that y is bounded. Hint: Note that for any fixed $s > 0$, e^{sx} blows up as $x \to +\infty$ and e^{-sx} blows up as $x \to -\infty$.

***Exercise* 6.5.101:** *Solve*

$$\begin{aligned} &y_t + y_x = 1, \qquad 0 < x < \infty, \ \ t > 0, \\ &y(0,t) = 0, \quad y(x,0) = 0. \end{aligned}$$

***Exercise* 6.5.102:** *For a $c > 0$, solve*

$$\begin{aligned} &y_t + y_x + cy = 0, \qquad 0 < x < \infty, \ \ t > 0, \\ &y(0,t) = \sin(t), \quad y(x,0) = 0. \end{aligned}$$

Exercise* 6.5.103*: *Find the corresponding ODE problem for $Y(x)$, after transforming the t variable*

$$y_{tt} + 3y_{xx} + y = x + t, \qquad -1 < x < 1, \ \ t > 0,$$
$$y(-1,t) = 0, \quad y(1,t) = 0, \quad y(x,0) = (1-x^2), \quad y_t(x,0) = 0.$$

Do not solve the problem.

Exercise* 6.5.104*: *Use the Laplace transform in t to solve*

$$y_{tt} = y_{xx}, \qquad -\infty < x < \infty, \ \ t > 0,$$
$$y(x,0) = \sin(x), \quad y_t(x,0) = 0.$$

Assume that y is bounded. Hint: Note that for any fixed $s > 0$, e^{sx} blows up as $x \to +\infty$ and e^{-sx} blows up as $x \to -\infty$.

Exercise* 6.5.105*: *Use the Laplace transform in t to solve*

$$y_t = y_{xx}, \qquad 0 < x < \infty, \ \ t > 0,$$
$$y(0,t) = f(t), \quad y(x,0) = 0,$$

where $f(t)$ is some function. Assume that y is bounded. Give the answer as a convolution.